

Get rewarded for giving your data away? The rise of ‘neobrowsers’

Targeting and profiling of individuals with the help of adtech is now turned around to consumers’ benefit. **Juliette Faivre** reports on Gener8 and Brave.

The abuse of power by advertisers, and ever-more sophisticated profiling, has led to multiple privacy scandals in the last few years. In Europe, lawmakers have adopted regulations to draw a line between acceptable and unacceptable advertising behaviours. The EU GDPR (2016), and Privacy and Electronic Communications Directive (2002) have led to landmark shifts in the sphere of data protection, especially for those firms profiting from personalised, yet sometimes intrusive customer experiences. Adtech is watched closely by regulators, as recent examples by Belgian¹, Irish, Italian and Spanish DPAs show². The UK ICO has also done much work in this area - it issued an opinion on adtech in November 2021 ([PL&B UK Report January 2022, p.14](#)) and its investigations continue.³

A SHIFTING PARADIGM FOR ONLINE ADVERTISING: THE RISE OF “NEO-BROWSERS”

Alongside regulation, new browsers claiming privacy and data control have known a significant rise in popularity among Internet users, who increasingly care about their data. As Sam Jones, Founder of Gener8, stated in an interview with Forbes, “We mindlessly click ‘accept’ on the cookies pop-up that gets in the way of the article we’re trying to read, but the reality is [...] thousands of companies potentially start tracking, sharing, and profiting from your online activity. But this is our information, it’s our data, and under GDPR legislation, legally our property”.⁴

Headlines such as “advertisers have gained more than \$500 billion from tracking and selling data last year [2021]”⁵, have alerted commentators to “neo-browsers” that are promoting attractive and socially-responsible stakeholders’ values which can, or should be, reflected in the AdTech process.

Moreover, “as the corporate use of personal data has grown over the years,

so has the interest in the value of data”⁶, encouraging users to enter the debate of personal data used for advertising, while giving birth to the field of research of infonomics (discipline of asserting economic value to data assets).

As a consequence of this emerging phenomenon, start-ups such as Gener8 (UK) and Brave (US) among others, have strived to work towards a similar mission, which is one of taking back control of user’s data. Both companies offer a downloadable, standalone browser with a “no-reward” default setting or a “reward” mode. For the latter, explicit consent to share data with the browser is required from the side of the user.

It is worth pointing out that despite the similarities, these two neobrowsers operate in different ways.

GENER8 REWARDS: THE FOCUS ON SHARED VALUE WITH DATA MONETISATION

In the “reward mode”, Gener8 will “monetise your data on your behalf: it will tailor adverts that you see, anonymise your data and use it for research”⁷. It will act as an ad blocker for “traditional” browsing, by replacing most web ads with the pool of content it receives from its advertising partners. Therefore, drawing on user preferences, personalised advertising, surveys and smart tracking will be collected, while ensuring to anonymise data with a randomly generated alphanumeric identifier (see Terms of Use⁸). Consequently, Gener8 will reward users with points that they can exchange for products (e.g. Alexa, Playstation) or vouchers (e.g. Amazon cards) in the marketplace. According to Sam Jones, 80% of the ad revenue goes to the user, whilst Gener8 takes a 20% cut. The average user should expect to redeem £5-£25 worth of points every month as a ‘reward’ for their browsing.⁹

While Gener8’s business model has been heavily praised by investors following its appearance in the investor show Dragon’s Den, the company has nonetheless received criticisms as to the fairness of the ad replacement model. In fact, small publishers whose website experiences are altered by ad blockers are not able to earn from ad displays. This creates a major gap with larger creators who are able to sponsor services such as Gener8, for paid ad display (or who simply have other sources of revenue). Moreover, another point that could be raised, is the validity of its behavioural insights, in case the browser is used by multiple people who demonstrate inconsistent preferences over time.

Gener8’s privacy policy is exhaustive on data collection and retention. In fact, the company reserves the rights to “collect preferences, browsing history and purchases made” under the reward mode, while ensuring that the sensitive data is not associated with the user profile containing sensitive, personal identity (using the system of pseudonymisation, through anonymous keys). This does not allow the company or its partners to target individuals intentionally. While the privacy policy does not exactly communicate the data retention timeframe (“We will only keep your data for as long as we need to.”¹⁰), it ensures the permanent account and data deletion if a user has not used the Gener8 services for over five years. In terms of data sharing, Gener8 focuses on transparency by disclosing a full list of advertising partners and third parties on its website. The company reveals that while being based in the UK, its operations reach beyond Europe; however it guarantees data “receives additional protection as required by law” when transferred.

Moreover, in accordance with guidance by the Information Commissioner’s Office (ICO), Gener8 users are guaranteed the rights to access, correct,

erase, restrict, transfer, object and withdraw any consent and data, though these “may not always be available” due to contextual circumstances.

Brave: A SOLUTION WITH AN EMPHASIS ON PRIVACY

Brave, headquartered in San Francisco (California), operates under a different revenue model than Gener8, and the company does not sell nor use personal data for research, with a much greater focus on privacy. The company ensures that “your data [personal, behaviour activity, browsing history] never leaves your device,”¹¹ as it solely rewards users for attention in the traditional sense of advertising – displays of rotational, yet personalised billboards. In fact, users activate Brave images, which appear in the browser’s background or they can choose to opt-in to respectful push notifications (upon consent). These ads come from Brave Verified Advertisers, and are matched to the individual’s preferences locally – only based on the browser’s analysis.

While Brave does send performance feedback on ad campaigns (based on an aggregation of user contributions), these do not contain individual user values (even if codified) and are solely used for confirmation of activity. Encryption keys and other systems ensure Brave’s servers do not access the data, and they prevent brands from tracking user activity.¹² This engagement relates to the “anonymous-but-accountable”¹³ campaign reporting scheme.

With a user interface that can be personalised, the flexibility offered by Brave is greater than Gener8’s, which allows users to control the frequency and programming of ads, in addition to the quantity of shared information with the browser.

Similarly to Gener8, users gain the browser’s native currency – Basic Attention Tokens (BATs) – in exchange for their attention, which are utility tokens based on Ethereum technology. These are stored in a user wallet (“Uphold”, Brave’s third-party) and they can be exchanged for cryptocurrencies, monetary vouchers or even donated to content creators (Brave Verified Publishers, BVP). As a difference from Gener8, BVPs (usually small creators who are disadvantaged by ad

blockers), can receive contributions from Brave users who visit their website, supporting their revenue inflow.

A downside from earning rewards with Brave is the much slower pace, approximately 3-4 BATs per month for the average user, which has a conversion rate of 1 BAT = US\$0.34 at the current date of writing.¹⁴ Usually, the user can expect to receive up to 70% share of the price shared to the advertiser (worth in BAT), per impression.¹⁵

Brave’s EDGE IN DIFFERENTIAL PRIVACY: EVER-EVOLVING SAFETY FEATURES

As a means to preserve online privacy, Gener8 and Brave use pseudonymization techniques to mask user identities (through user “identifiers” – usually composed of letters and numbers). It is important to mention that while these techniques may be used with good intent, the UK ICO stated that this solution was usually “focused on reducing direct identifiability”, leaving space for potential indirect identifiability (opinion piece published in November 2021). In fact, the ICO claims that “effective anonymisation requires organisations to demonstrate how to mitigate identifiability risk, and if it is reasonably likely to identify an individual directly or indirectly – then the data is not anonymised”¹⁶. Moreover, it is essential to note that hackers and malicious parties could use “re-identification” techniques or be able to trace back the data to its origins, according to Heather Devane, Content Marketing Manager at Immuta (healthcare company)¹⁷.

Aware of these shortcomings, Brave Research recently published a post on its blog of privacy feature updates (July 2022)¹⁸. It described an innovative implementation of “k-anonymisation”, under a newly-developed system called STAR. K-anonymization is a computational method to further enhance privacy, by which only “popular” values will be displayed to the data collector, and the “unique” values will be hidden. In practice, this allows for preferences generalisation based on behavioural insights. Following Brave’s example of ice cream flavours, if the three most popular options are chocolate, vanilla and strawberry, a user who’s favourite flavour is “olive” will be hidden from

the data even though all users are anonymous (through identifiers). This is because the olive flavour, as a unique value, could be traced back to the user indirectly, in aggregation with other data points. This process is known as ‘Hiding outliers into the crowd’ (according to a paper by Wang, 2015¹⁹), and k-anonymity is the “property that indicates the re-identifiability potential of its records”²⁰.

Because of the method’s costly implementation, decision-making limitations (e.g. the assessment and contextual definition of an outlier) and other technical complexities, Brave’s researchers developed the open-source STAR system²¹ which solves these shortcomings while still ensuring a strong privacy protection. The company has mentioned that this revolutionary system, would be used in current and future Brave products, also presented in global conferences²² and discussed, to possibly be standardised in the Internet Engineering Task Force (IETF) network. In fact, standardising such a system would ban intrusive and potentially dangerous microtargeting of users, which in the hands of malicious parties, could be used to harm individuals.

THE IMPORTANCE OF TRUST IN A DIGITAL WORLD

Companies should have the moral obligation to build a protective digital sphere that aligns with human-centered and sustainable values. While financial metrics remain crucial to the performance of companies, it is important to note that environmental, social, and governance (ESG) impacts are increasingly examined – and companies need to make sure their action plan agendas align accordingly.

Rachel Botsman, author of the book ‘Who Can You Trust?’, mentions how companies act as ‘trust engineers’²³ in our society, bridging gaps between the real and online world. There is increasing complexity of information technology and its related systems, and companies, therefore, need to take ethical engagements such as maintaining an intelligible discourse about their provisions concerning data privacy, in order to restore or maintain a healthy relationship with the ever-expanding online community.

As discussed, neobrowsers are working towards a solid basis for increased accountability and shared value with consumers; a model that should be further supported by the regulation landscape. While they still present flaws, the neobrowsers in BETA

versions will undergo multiple iterations in the near future, attracting more users over time. Should traditional browsers feel threatened, by a non-regulatory force that may impact viability in the long-term?

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IAB Europe case referred to the CJEU

The Brussels Appeal Court has referred the IAB Europe (Interactive Advertising Bureau) case, concerning IAB's Transparency and Consent Framework, to the Court of Justice of the European Union (CJEU).

Belgium's Data Protection Authority says that the IAB framework is in breach of the GDPR, and that the organization is a (joint) data controller for processing operations performed

by the users of the standard, i.e., publishers and adtech vendors.

Law firm Covington, representing IAB Europe, reports:

"The Market Court [of the Brussels Appeal Court] decided that the Supervisory Authority violated the principle of due care in investigating the facts of the case and the allegations brought by the complainants. The court supported the claim made by IAB Europe that the

Supervisory Authority just took the allegations for granted without actually investigating them itself. As a result, the court found its decision was not based on proven objective facts."

A decision by the CJEU is not expected for at least two years.

• See www.insideprivacy.com/advertising-marketing/brussels-appeal-court-refers-iab-europe-case-to-cjeu/



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Proposed US federal data privacy law offers strong protections but only to US residents

Professor Graham Greenleaf analyses the US Data Privacy and Protection Act from a global perspective.

To the surprise of most observers, businesses and privacy advocates, the United States is closer than it has ever been to enacting a national data privacy law for its private sector, the American

Data Privacy and Protection Act (ADPPA).¹ The proposed Act has strong bipartisan support, as shown in a 53-2 vote in its favour by the US

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G7 countries work towards elements of convergence

Bridging differences, and emphasising democratic values are key elements for promoting Data Free Flow with Trust.

By **Dr. Heiko Haupt** and **Georg Witos** of Germany's Federal Data Protection Commission.

The Data Protection Authorities of the G7 countries; Canada, France, Germany, Italy, Japan, the United Kingdom, the United States and the European Union, met in Germany 7-8 September to share

knowledge about the prospects for "international data spaces".

The meeting, convened under the chairmanship of the German Federal

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